

Qualification Pack



Internet of Things (IoT) Developer

QP Code: NIE/ELE/Q0503

Version: 1.0

NSQF Level: 4.5

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Qualification Pack

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NIE/ELE/Q0503: Internet of Things (IoT) Developer

Brief Job Description

The Internet of Things (IoT) Developer qualification enables the participants to develop and deploy IoT based products. After course completion, the trained learner will exhibit the role of Programmer cum Developer to design IoT based software and hardware Industry based solutions. Hands-on expertise in the areas of Python programming language along with in-depth handling of Raspberry Pi are the takeaways of this course. This course enhances the working of Microcontroller based devices in interdisciplinary areas without degrading their performance.

Personal Attributes

participants should have the ability to analyze and solve problems, ability to deploy IoT solution as per requirement

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ELE/N0506: Identification and troubleshooting of Basic Electronics components](#)
2. [NIE/ELE/N0515: Conceptualising IoT Platform - Arduino](#)
3. [NIE/ELE/N0514: Conceptualising IoT based use-cases](#)
4. [NIE/ELE/N0504: Fundamentals of wireless IoT using NodeMCU](#)
5. [NIE/ELE/N0505: Conceptualising Single Board Computer as an IoT Platform Raspberry Pi](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	IoT Solution, Industrial IoT (IIoT)
Country	India
NSQF Level	4.5

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Credits	18
Aligned to NCO/ISCO/ISIC Code	QG-4.5-IT-00346-2023-V1-NIELIT
Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) with NA of experience OR Pursuing 1st year of UG (Pursuing 1st year of UG and continuous education) with NA of experience OR Pursuing 3rd year of 3-year diploma after 10th (Pursuing 3rd year of 3-year diploma after 10th and continuous education) with NA of experience OR Completed 3 year diploma after 10th with NA of experience OR Completed 2nd year diploma after 12th (Completed 2nd year of 2-year diploma after 12th) with NA of experience OR Pursuing 2nd year of 2-year diploma after 12th (Pursuing 2nd year of 2- year diploma after 12 and continuous education) with NA of experience OR Previous relevant Qualification of NSQF Level (Previous relevant Qualification of NSQF Level 4 and with minimum education as 8 th Grade pass) with NA of experience 1.5 Years
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	no
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	26/03/2026
NSQC Approval Date	26/03/2023
Version	1.0
Reference code on NQR	QG-4.5-IT-00346-2023-V1-NIELIT
NQR Version	1

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NIE/ELE/N0506: Identification and troubleshooting of Basic Electronics components

Description

The NOS covers examine and identification of several electronic devices and components

Scope

The scope covers the following :

- Identification of electronics components
- Understanding the Fundamentals of basic electronics
- Troubleshooting of electronics components
- Understanding the operation of measurement devices.

Elements and Performance Criteria

Identification of electronics components

To be competent, the user/individual on the job must be able to:

PC1. Knowledge of basic electrical and electronic components

PC2. able to identify and distinguish electronic components

Fundamentals of basic electronics

To be competent, the user/individual on the job must be able to:

PC3. knowledge of basic electronics and electronic devices

PC4. Troubleshooting of electronics components

Understanding the operation of measurement devices.

To be competent, the user/individual on the job must be able to:

PC5. Knowledge of electrical and electronic instrumentation

PC6. able to understand how to operate the multimeter, ammeter, voltmeter, voltage supply etc

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identification of electronics components</i>	10	5	-	-
PC1. Knowledge of basic electrical and electronic components	-	-	-	-
PC2. able to identify and distinguish electronic components	-	-	-	-
<i>Fundamentals of basic electronics</i>	10	5	-	-
PC3. knowledge of basic electronics and electronic devices	-	-	-	-
PC4. Troubleshooting of electronics components	-	-	-	-
<i>Understanding the operation of measurement devices.</i>	8	4	-	-
PC5. Knowledge of electrical and electronic instrumentation	-	-	-	-
PC6. able to understand how to operate the multimeter, ammeter, voltmeter, voltage supply etc	-	-	-	-
NOS Total	28	14	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0506
NOS Name	Identification and troubleshooting of Basic Electronics components
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	IoT Solution, IoT Solution
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	26/03/2023
Next Review Date	26/03/2026
NSQC Clearance Date	26/03/2023

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NIE/ELE/N0515: Conceptualising IoT Platform - Arduino

Description

The NOS covers IoT development boards, programming and Interfacing

Scope

The scope covers the following :

- Introduction to microprocessor and microcontroller
- Introduction to Internet of Things(IoT) applications, protocols, usecases
- Introduction to Arduino
- Embedded C Language
- Interfacing of sensors and actuators with Arduino Boards

Elements and Performance Criteria

Introduction to microprocessor and micro controller

To be competent, the user/individual on the job must be able to:

PC1. Knowledge of microprocessors and microcontrollers

PC2. knowledge of IoT development boards

Introduction to Internet of Things(IoT)

To be competent, the user/individual on the job must be able to:

PC3. knowledge of IoT architecture, building blocks and use cases

Introduction to C programming

To be competent, the user/individual on the job must be able to:

PC4. able to work with C and Embedded C

Arduino as IoT development board

To be competent, the user/individual on the job must be able to:

PC5. Knowledge of Arduino boards

PC6. Able to interface sensors and actuators on Arduino boards for IoT applications

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to microprocessor and micro controller</i>	10	-	-	-
PC1. Knowledge of microprocessors and microcontrollers	-	-	-	-
PC2. knowledge of IoT development boards	-	-	-	-
<i>Introduction to Internet of Things(IoT)</i>	15	-	-	-
PC3. knowledge of IoT architecture, building blocks and use cases	-	-	-	-
<i>Introduction to C programming</i>	20	12	5	-
PC4. able to work with C and Embedded C	-	-	-	-
<i>Arduino as IoT development board</i>	22	16	5	-
PC5. Knowledge of Arduino boards	-	-	-	-
PC6. Able to interface sensors and actuators on Arduino boards for IoT applications	-	-	-	-
NOS Total	67	28	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0515
NOS Name	Conceptualising IoT Platform - Arduino
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	IoT Solution, IoT Solution
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	26/03/2023
Next Review Date	26/03/2026
NSQC Clearance Date	26/03/2023

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NIE/ELE/N0514: Conceptualising IoT based use-cases

Description

The NOS covers various IoT based use case and prototype development

Scope

The scope covers the following :

- Smart Street Light control
- Home automation using Arduino
- Password enabled Digital Lock using Arduino

Elements and Performance Criteria

Smart Street Light control

To be competent, the user/individual on the job must be able to:

PC1. IoT based prototype development for Smart Street Light control

Home automation using Arduino

To be competent, the user/individual on the job must be able to:

PC2. Development of a home automation system

Password enabled Digital Lock using Arduino

To be competent, the user/individual on the job must be able to:

PC3. Able to develop Digital Locks using Arduino

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Smart Street Light control</i>	9	4	-	-
PC1. IoT based prototype development for Smart Street Light control	-	-	-	-
<i>Home automation using Arduino</i>	10	5	-	-
PC2. Development of a home automation system	-	-	-	-
<i>Password enabled Digital Lock using Arduino</i>	10	5	-	-
PC3. Able to develop Digital Locks using Arduino	-	-	-	-
NOS Total	29	14	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0514
NOS Name	Conceptualising IoT based use-cases
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	IoT Solution, IoT Solution
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	26/03/2023
Next Review Date	26/03/2026
NSQC Clearance Date	26/03/2023

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NIE/ELE/N0504: Fundamentals of wireless IoT using NodeMCU

Description

This NOS covers development of IoT applications using wireless modules

Scope

The scope covers the following :

- Fundamentals and architecture of wireless IoT
- TCP/IP modelling for IoT
- NodeMCU as an IoT Platform
- Fundamentals of cloud Platforms
- Implementation of cloud based IoT use cases

Elements and Performance Criteria

Fundamentals and architecture of wireless IoT

To be competent, the user/individual on the job must be able to:

PC1. Knowledge of Web stack and IoT stack

PC2. Knowledge of lightweight protocols

NodeMCU as an IoT Platform

To be competent, the user/individual on the job must be able to:

PC3. Working with NodeMCU and related development boards

Implementation of cloud based IoT use cases

To be competent, the user/individual on the job must be able to:

PC4. understanding of cloud and cloud platforms for IoT

PC5. Monitoring and controlling through cloud platforms in an IoT ecosystem

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamentals and architecture of wireless IoT</i>	13	-	-	-
PC1. Knowledge of Web stack and IoT stack	-	-	-	-
PC2. Knowledge of lightweight protocols	-	-	-	-
<i>NodeMCU as an IoT Platform</i>	15	10	5	-
PC3. Working with NodeMCU and related development boards	-	-	-	-
<i>Implementation of cloud based IoT use cases</i>	15	10	5	-
PC4. understanding of cloud and cloud platforms for IoT	-	-	-	-
PC5. Monitoring and controlling through cloud platforms in an IoT ecosystem	-	-	-	-
NOS Total	43	20	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0504
NOS Name	Fundamentals of wireless IoT using NodeMCU
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	IoT Solution, IoT Solution
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	26/03/2023
Next Review Date	26/03/2026
NSQC Clearance Date	26/03/2023

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NIE/ELE/N0505: Conceptualising Single Board Computer as an IoT Platform Raspberry Pi

Description

NOS covers raspberry pi as an IoT platform and Implementation of IoT based use cases using Raspberry Pi

Scope

The scope covers the following :

- Introduction to Raspberry Pi
- Basics of Python Programming
- Interfacing of sensors and actuators with Raspberry Pi
- Implementation of IoT based use cases using Raspberry Pi

Elements and Performance Criteria

Introduction to Raspberry Pi

To be competent, the user/individual on the job must be able to:

PC1. Knowledge of raspberry pi, botting procedure and pin configuration

Basics of Python Programming

To be competent, the user/individual on the job must be able to:

PC2. Able to write codes in python for IoT

Implementation of IoT based use cases using Raspberry Pi

To be competent, the user/individual on the job must be able to:

PC3. able to Interface and program sensors and actuators with Raspberry Pi for IoT applications

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Raspberry Pi</i>	8	-	-	-
PC1. Knowledge of raspberry pi, botting procedure and pin configuration	-	-	-	-
<i>Basics of Python Programming</i>	15	7	5	-
PC2. Able to write codes in python for IoT	-	-	-	-
<i>Implementation of IoT based use cases using Raspberry Pi</i>	10	7	5	-
PC3. able to Interface and program sensors and actuators with Raspberry Pi for IoT applications	-	-	-	-
NOS Total	33	14	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0505
NOS Name	Conceptualising Single Board Computer as an IoT Platform Raspberry Pi
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	IoT Solution, IoT Solution
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	26/03/2023
Next Review Date	26/03/2026
NSQC Clearance Date	26/03/2023

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.

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6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0506.Identification and troubleshooting of Basic Electronics components	28	14	-	-	42	12
NIE/ELE/N0515.Conceptualising IoT Platform - Arduino	67	28	10	-	105	30
NIE/ELE/N0514.Conceptualising IoT based use-cases	29	14	-	-	43	12
NIE/ELE/N0504.Fundamentals of wireless IoT using NodeMCU	43	20	10	-	73	21
NIE/ELE/N0505.Conceptualising Single Board Computer as an IoT Platform Raspberry Pi	33	14	10	-	57	16
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	9
Total	220	120	30	-	370	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.